OFFICIAL BIDDING DOCUMENT

Telecommunications Technology Solutions Inc. (TelTech)

Corporate Tender Secretariat · Procurement & Network Engineering Division

RFP-TELTECH-FWA-2026-001 / APPENDIX A

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Technical Specifications & Compliance Matrix

Mandatory Technical Requirements for RAN, Power, 5GC, OSS/BSS, RF, Managed Services & CPE

DOCUMENT NUMBER	AUTHOR	TARGET SPECTRUM	BIDDING STATUS
RFP-TELTECH-FWA-2026-001	OLIVER TUNGOL	Band n50 (1.5 GHz, 60 MHz TDD)	Official Request for Proposal

Instructions to Bidders: Complete Column D with "C" (Full Compliance), "M" (Minor Deviation - explain in remarks), or "NC" (Non-Compliant). Bidders must provide specific technical values and document references.

SECTION 1: RADIO ACCESS NETWORK (RAN) TECHNICAL SPECIFICATIONS

ITEM ID	SUB-SYSTEM / FEATURE	DETAILED SPECIFICATION REQUIREMENT	COMPLIANCE (C / M / NC)	BIDDER PROPOSED SPECIFICATION & REFERENCE
RAN-01	3GPP Standard	3GPP Release 16 and Release 17 compliant 5G NR Standalone (SA) architecture.		
RAN-02	Frequency Band	Band n50 (1432 – 1517 MHz, 1.5 GHz L-Band TDD). Full RF support for 1432–1517 MHz.		
RAN-03	Channel Bandwidth	Continuous 60 MHz channel bandwidth on a single carrier; software-configurable to 40 MHz / 50 MHz / 60 MHz.		
RAN-04	Duplex Mode	Time Division Duplex (TDD) with flexible frame configuration (e.g., 4:1 DL:UL ratio with special slot 10:2:2).		
RAN-05	Subcarrier Spacing	30 kHz Subcarrier Spacing (SCS) with normal cyclic prefix (CP).		
RAN-06	Synchronization	IEEE 1588v2 Precision Time Protocol (PTP G.8275.1 profile), Synchronous Ethernet (SyncE), and GNSS (GPS/BeiDou/GLONASS) receiver onboard BBU.		
RAN-07	Radio Configuration	Standard macro site: 3 sectors, 8T8R configuration (3x 8T8R RRU or integrated AAU per site).		
RAN-08	RF Output Power	Minimum 8x 40W (320W total RF output power per sector/RRU). Digital power control with dynamic power backoff.		
RAN-09	Beamforming	Digital + analog beamforming (horizontal and vertical); dynamic Multi-User MIMO (MU-MIMO) pairing in DL and UL.		
RAN-10	Modulation Schemes	Downlink: Up to 256-QAM adaptive modulation. Uplink: Up to 64-QAM / 256-QAM adaptive modulation.		

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RAN-11	MIMO Support	Downlink: Up to 8-layer transmission across multiple users (MU-MIMO); up to 4-layer transmission to single 4R CPE. Uplink: Up to 2-layer single user MIMO; 4-layer MU-MIMO.		
RAN-12	Sector Throughput	Aggregated single-sector peak throughput >= 450 Mbps in 60 MHz DL under 4:1 TDD frame configuration.		
RAN-13	Subscriber Capacity	BBU and sector software must support >= 350 active concurrent RRC connected CPEs per sector (>= 1,000 per 3-sector site) without degradation.		
RAN-14	BBU Architecture	Modular 19" rack-mountable subrack (<= 3U height). Dual hot-swappable -48V DC power supplies. Redundant main control units.		
RAN-15	Fronthaul Interfaces	Minimum 6x 25GE eCPRI optical interfaces per BBU; support for fiber-saving BiDi transceivers.		
RAN-16	Backhaul Interfaces	Minimum 2x 10GE + 2x 25GE SFP28 interfaces supporting IP routing, VLAN tagging (802.1Q), and Segment Routing / IP MPLS.		
RAN-17	Environmental Rating	Outdoor RRU/AAU: IP65 / IP67 ingress protection; operating temperature -40°C to +55°C; wind load resistance up to 250 km/h.		
RAN-18	Energy Saving Modes	Micro-sleep, symbol shutdown, deep sleep during low-traffic periods, reducing idle power consumption by >= 40%.		
RAN-19	Antenna Specifications	8-port cross-pol antenna, 65° horizontal HPBW, >= 17.5 dBi gain, integrated AISG 2.0 internal remote electrical tilt (eRET) adjustable 2° to 12°.		
RAN-20	PIM Performance	Passive Intermodulation (PIM) <= -153dBc @ 2 x 43 dBm carrier test.		

SECTION 2: POWER SYSTEMS & PASSIVE SITE INFRASTRUCTURE

ITEM ID	SUB-SYSTEM / FEATURE	DETAILED SPECIFICATION REQUIREMENT	COMPLIANCE (C / M / NC)	BIDDER PROPOSED SPECIFICATION & REFERENCE
PWR-01	DC Power Plant	19" 3U/4U modular high-efficiency switchmode rectifier system; continuous rating >= 6.0 kW @ -48V DC.		
PWR-02	Rectifier Modules	Hot-swappable 2.0 kW or 3.0 kW modules in N+1 redundant configuration. Operating efficiency >= 96.0%.		
PWR-03	Input Voltage Range	Wide AC input operating range: 85V AC to 300V AC nominal single phase (230V, 60 Hz), withstanding rural grid fluctuations.		
PWR-04	Lithium Battery (BESS)	Lithium Iron Phosphate (LiFePO4 / LFP) battery energy storage system: Sized to deliver at least 12.0 kWh usable energy to 3.0 kW DC continuous BTS load over 4.0 continuous hours without AC power, dimensioned for max 80% usable DoD and 80% end-of-life SoH retained capacity (minimum >= 20.0 kWh nominal installed capacity, minimum 400 Ah to 450 Ah @ 48V, dimensioned via mandatory loss-based engineering calculation delivering 12.0 kWh net usable energy at 80% DoD and 80% EOL SoH). Bidder shall submit formal load-autonomy calculation sheet, manufacturer discharge curves, and guarantee 4.0h autonomy at SAT.		

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PWR-05	Battery Autonomy	Guaranteed >= 4.0 hours continuous operation at 3.0 kW continuous BTS load without AC utility input, verified by full discharge test at site acceptance.		
PWR-06	Battery Cycle Life	>= 4,000 cycles at 80% Depth of Discharge (DoD) at +25°C; built-in intelligent Battery Management System (BMS).		
PWR-07	Standby Genset	For Greenfield (Class T4) sites: 15 kVA / 12 kW prime, 230V, 60 Hz diesel generator in sound-attenuated weatherproof canopy (<68dBA @ 7m).		
PWR-08	Fuel Autonomy	Integrated sub-base fuel tank with minimum 500L–1,000L capacity providing >= 72hours continuous runtime under full load.		
PWR-09	Automatic Transfer (ATS)	Microprocessor-controlled ATS with 4-pole motorized breaker and mechanical interlock, transferring power within 30 seconds of grid failure.		
PWR-10	Surge Protection	Coordinated SPD Type 1+2 on AC input (>= 40kA) and DC distribution busbar (>= 20kA).		
PWR-11	TowerCo Tap-Off Kit	Standardized co-location DC tap box with integrated Class 1.0 revenue-grade kWh energy meter and RS485/Modbus telemetry.		
PWR-12	Grounding & Lightning	Comprehensive grounding system achieving <= 5.0 Ohms earth electrode resistance; copper busbars and copper-clad steel ground rods.		

SECTION 3: 5G CORE NETWORK (5GC SA) & DATACENTER INFRASTRUCTURE

ITEM ID	SUB-SYSTEM / FEATURE	DETAILED SPECIFICATION REQUIREMENT	COMPLIANCE (C / M / NC)	BIDDER PROPOSED SPECIFICATION & REFERENCE
5GC-01	Architecture	Cloud-native, containerized microservices 5G Standalone (5GC SA) compliant with 3GPP Release 16/17 Service-Based Architecture (SBA).		
5GC-02	Network Functions	Full NF suite: AMF, SMF, UPF, PCF, UDM, UDR, AUSF, NRF, NSSF, NEF, CHF, and SEPP.		
5GC-03	Subscriber Sizing	Day-1 licensed capacity: 1,000,000 registered FWA subscribers. Scalable in 500k increments up to 3,000,000 subscribers.		
5GC-04	Geo-Redundancy	Active-Active geo-redundant dual-datacenter deployment (Metro Manila Primary DC and Clark Pampanga DR DC).		
5GC-05	Session-Preserving Failover	Automatic subscriber session continuity (zero session loss across UDR/UDM and SMF clusters) and automated user-plane traffic restoration <= 30 seconds during datacenter failure or scheduled DR drill.		
5GC-06	UPF Throughput	UPF dimensioned for >= 80 Gbps usable wire-speed throughput per 2U dual-socket compute node with DPDK and SR-IOV acceleration.		
5GC-07	Carrier-Grade NAT	Integrated or inline CG-NAT supporting deterministic port allocation, logging per NTC legal intercept mandate, and 10:1 to 16:1 subscriber ratio.		
5GC-08	Deep Packet Inspection	Embedded DPI capable of Layer-7 application recognition, speed throttling per subscriber tariff tier, and fair-use policy enforcement.		

ITEM ID	SUB-SYSTEM / FEATURE	DETAILED SPECIFICATION REQUIREMENT	COMPLIANCE (C / M / NC)	BIDDER PROPOSED SPECIFICATION & REFERENCE
5GC-09	QoS & Slicing	5G QoS Flow mapping (5QI 1, 2, 7, 8, 9) and Network Slicing (S-NSSAI) separating consumer FWA, SME enterprise, and management traffic.		
5GC-10	Datacenter Compute	Enterprise 2U server nodes (Dual Intel Xeon Gold / AMD EPYC, 512GB RAM, redundant NVMe enterprise SSDs, dual 100GE NICs).		
5GC-11	Datacenter Fabric	Leaf-spine datacenter networking architecture (32 x 100GE spine switches, 48 x 25GE + 6 x 100GE leaf switches) with EVPN-VXLAN.		
5GC-12	Security & Scrubbing	Redundant carrier-grade perimeter next-generation firewalls (>= 100 Gbps throughput) and inline anti-DDoS scrubbing appliances.		

SECTION 4: CONVERGED DIGITAL OSS/BSS & FIELD FORCE AUTOMATION

ITEM ID	SUB-SYSTEM / FEATURE	DETAILED SPECIFICATION REQUIREMENT	COMPLIANCE (C / M / NC)	BIDDER PROPOSED SPECIFICATION & REFERENCE
BSS-01	Product Catalog & Tariffs	Configurable telecom product catalog supporting 4 consumer FWA plans (₱599, ₱799, ₱999, ₱1,299), SME tiers, and flexible speed-cap policies.		
BSS-02	Prepaid Charging (CHF)	Real-time convergent charging system (CCS/CHF) integrated via 3GPP Nchf/Diameter Ro, supporting real-time balance rating and session cutoff.		
BSS-03	Omnichannel Payments	Native REST API integration with Philippine payment gateways: GCash, Maya (PayMaya), 7-Eleven CLiQQ OTC, Bayad Center, and QR Ph.		
BSS-04	Subscriber Portal & App	Native Android/iOS subscriber self-care mobile app and responsive web portal for balance checking, online top-up, and speed-test diagnostics.		
OSS-01	Field Force App (FMS)	Mobile app for field technicians: Real-time work-order dispatch, turn-by-turn navigation, barcode/QR scan of CPE serials, and digital e-signature.		
OSS-02	RF Signal Verification	Mobile field tool must perform live outdoor RF signal validation (RSRP >= -95 dBm, SINR >= 10 dB) before allowing install closure.		
OSS-03	Dispatch Throughput	Algorithmic dispatch engine capable of dynamically optimizing and dispatching >= 3,500 installation and maintenance work orders daily.		
OSS-04	CPE Inventory & WMS	Serialized lifecycle tracking of CPE hardware across central warehouse, technician vehicle float, active subscriber homes, and repossession.		
OSS-05	Reverse Logistics	Reverse logistics workflow tracking recovered churned CPEs through testing, factory reset, cleaning, re-boxing, and return to stock (>= 70% target).		
OSS-06	Fault & Performance OSS	Unified multi-vendor EMS/NMS monitoring 5,000 RAN sites, transmission links, and 5G Core with automated RCA and alarm correlation.		
OSS-07	Regulatory Compliance	Automated generation of quarterly NTC Quality of Service (QoS) and broadband speed performance audit reports.		

SECTION 5: RF PLANNING, SITE SURVEY & CLUSTER OPTIMIZATION

ITEM ID	SUB-SYSTEM / FEATURE	DETAILED SPECIFICATION REQUIREMENT	COMPLIANCE (C / M / NC)	BIDDER PROPOSED SPECIFICATION & REFERENCE
RFP-01	Planning Tool	Carrier-grade RF planning tool (Atoll, Asset, Planet) with high-resolution digital terrain and clutter data (<= 5m resolution).		
RFP-02	Propagation Model	Calibrated Band n50 (1.5 GHz) 3D ray-tracing / standard propagation model calibrated against local Philippine terrain clutter.		
RFP-03	Nominal Siting & Audit	Siting validation against 13,746 tower inventory and Converge fiber network backbone routes (subject to verification on the ground); production of candidate search rings.		
RFP-04	Technical Site Survey	Comprehensive TSSR per site including tower structural load audit, wind-load calculation, panoramic drone photography (360°), and power audit.		
RFP-05	Coverage Target	Guaranteed outdoor RSRP >= -95 dBm and SINR >= 10 dB across >= 90% of residential settlement footprint in target barangay.		
RFP-06	Single Site Acceptance	Single Site Verification (SSV) stationary and drive tests verifying sector radiation, antenna RET, PCI, TAC, and max throughput.		
RFP-07	Cluster Tuning	Cluster optimization across 10–20 site clusters: Interference analysis, tilt/azimuth physical optimization, handover parameter tuning.		

SECTION 6: MANAGED SERVICES & 24/7 NETWORK OPERATIONS (MS)

ITEM ID	SUB-SYSTEM / FEATURE	DETAILED SPECIFICATION REQUIREMENT	COMPLIANCE (C / M / NC)	BIDDER PROPOSED SPECIFICATION & REFERENCE
MS-01	24/7 Tier-2/Tier-3 NOC	Continuous 24/7/365 active monitoring, incident triage, alarm dispatch, and escalation management across all network domains.		
MS-02	Network Availability	Overall end-to-end radio and core network service availability >= 99.70% monthly SLA.		
MS-03	Critical Incident MTTR	Mean Time to Restore full service for Critical / Severity-1 outages: <= 2hours in urban areas; <= 4hours in rural areas.		
MS-04	First Line Maint. (FLM)	Dedicated nationwide field dispatch teams with dedicated 4x4 vehicles, fusion splicers, OTDRs, spectrum analyzers, and climbing gear.		
MS-05	Preventive Maintenance	Quarterly scheduled site inspections: Antenna bracket torque checks, RF cable weatherproofing, battery impedance tests, genset runs.		
MS-06	Spare Parts Logistics	Automated SPMS maintaining regional buffer stock; guaranteed Next Business Day (NBD) hardware replacement delivery nationwide.		

SECTION 7: 5G CUSTOMER PREMISES EQUIPMENT (CPE - LOT 7)

ITEM ID	SUB-SYSTEM / FEATURE	DETAILED SPECIFICATION REQUIREMENT	COMPLIANCE (C / M / NC)	BIDDER PROPOSED SPECIFICATION & REFERENCE
CPE-01	3GPP Standards & Modes	3GPP Release 16/17 compliant 5G NR sub-6 GHz Standalone (SA) and Non-Standalone (NSA) operation.		
CPE-02	Frequency Bands	Native Band n50 (1432 – 1517 MHz, 60 MHz channel bandwidth) plus fallback sub-6 GHz bands (n1, n3, n7, n28, n41, n77, n78).		
CPE-03	Indoor CPE Antenna & MIMO	High-gain omnidirectional multi-element internal antenna array (minimum >= 8.0 dBi gain across Band n50); 4x4 MIMO DL, 2x2 MIMO UL.		
CPE-04	Outdoor CPE Antenna & MIMO	High-gain directional integrated antenna array (minimum >= 12.0 dBi gain across Band n50); 4x4 MIMO DL, 2x2 MIMO UL; adjustable pole/wall mounting bracket.		
CPE-05	Wi-Fi Subsystem	Dual-band Wi-Fi 6 (802.11ax) 2.4 GHz (2x2) and 5 GHz (2x2) concurrent, OFDMA, MU-MIMO, WPA3-Personal encryption.		
CPE-06	Ethernet Interfaces	Indoor CPE: Minimum 1x 2.5GE + 2x 1GE RJ-45 LAN ports. Outdoor CPE: 1x 2.5GE PoE input port plus companion indoor Wi-Fi 6 router/gateway and 802.3at PoE injector included in kit.		
CPE-07	Environmental Rating	Outdoor CPE: IP67 dust/water ingress protection, operating temperature -10°C to +55°C, 200 km/h wind survivability. Indoor CPE: IP20, 0°C to +40°C.		
CPE-08	Remote Device Management	Broadband Forum TR-069 and TR-369 (USP) support with complete TR-181 data model, SNMPv2/v3, and secure ACS provisioning.		
CPE-09	Firmware & Security	Secure Boot with cryptographic signature verification, automated FOTA (Firmware-Over-The-Air) batch upgrades, and operator lock capability.		
CPE-10	SIM / eSIM Support	Standard Nano-SIM (4FF) slot with anti-theft locking cover and optional GSMA-compliant eSIM MFF2 footprint.		
CPE-11	Power Supply	Indoor CPE: 230V AC / 12V DC external adapter (Philippine standard Type A/B plug). Outdoor CPE: 2.5GE-capable 802.3at/bt PoE power injector and 2.5GE WAN indoor gateway companion included.		
CPE-12	Certification & Warranty	NTC Type Approval compliance, CE/FCC certification, and minimum 24-month hardware replacement warranty (RMA <= 5 business days).		

(End of Appendix A — Technical Specifications & Compliance Matrix)